

# PERN ERP System Analysis

Comprehensive Technical Specification, Architectural Blueprint & Business Workflow

Stack: PostgreSQL • Express.js • React.js • Node.js

Document Version: 1.0 (Production Master)

## 1. Project Overview

The **PERN ERP (Enterprise Resource Planning)** project is a web-based management system developed using the full-stack **PERN Stack** (PostgreSQL, Express.js, React.js, and Node.js). The system is designed to streamline and automate the entire sales and inventory lifecycle of modern manufacturing and trading enterprises. It empowers organizations to seamlessly manage customer enquiries, generate automated price quotations, convert accepted offers into sales orders, dynamically reserve stock, and execute dispatches while guaranteeing multi-table transactional consistency and high-precision inventory tracking.

## 2. Problem Statement

In many traditional manufacturing and trading enterprises, sales operations, order processing, and inventory tracking are handled manually through disconnected spreadsheets, paper records, or fragmented single-purpose applications. This decentralized operational model introduces critical bottlenecks:

### Key Operational Inefficiencies Addressed:

- **Duplicate & Conflicting Orders:** Absence of real-time multi-user concurrency controls causes multi-allocation of identical stock items.
- **Inventory Mismatch:** Physical warehouse counts frequently diverge from recorded stock due to delayed manual logs.
- **Manual Calculation Errors:** Human errors during tax, discount, and total quotation price computations.
- **Poor Pipeline Visibility:** Inability to trace the origin of orders back to initial customer inquiries or underlying sales quotations.
- **Overselling Risk:** Absence of inventory pre-reservation mechanisms during order placement.

The PERN ERP application resolves these challenges by uniting all commercial, operational, and inventory workflows within a centralized, transactional database system.

## 3. Project Objectives

### Primary Objectives

- Centralize customer enquiry logging & tracking.
- Automate quotation generation with tax/discount calculation.
- Seamlessly transform approved quotations into Sales Orders.
- Implement automated real-time inventory reservation.

- Process order dispatches with direct physical stock updates.
- Maintain ACID-compliant database integrity across transactions.

### Secondary Objectives

- Implement state-of-the-art JWT authentication.
- Enforce strict Role-Based Access Control (RBAC).
- Expose RESTful, secure, and sanitized API endpoints.
- Provide visual analytics and operational activity logs.
- Build a highly responsive user interface with dynamic state handling.

## 4. Technology Stack Architecture

### P - PostgreSQL

#### Relational Database

- ACID Compliance for multi-table transactions.
- Row-level locking for inventory concurrency.
- Stores Users, Products, Orders, and Stock logs.

### E - Express.js

#### Backend Framework

- RESTful API route middleware.
- Request validation & error handling.
- Integrates JWT & bcrypt security layers.

### R - React.js

#### Frontend UI Architecture

- Dynamic Component reusability.
- Powered by Vite for sub-second builds.
- Axios for asynchronous API transport.

### N - Node.js

#### Server Runtime

- Non-blocking, event-driven I/O engine.
- Sequelize ORM abstraction.
- High throughput enterprise request handling.

## 5. System Architecture Diagram



## 6. User Roles & Permission Matrix

| Role       | Allowed Capabilities                                                                                                                                                                                                                       | System Restrictions                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADMIN      | <ul style="list-style-type: none"><li>• Full User &amp; Product CRUD operations.</li><li>• Confirm pending sales orders.</li><li>• Execute stock reservation &amp; dispatches.</li><li>• Access global audit logs &amp; reports.</li></ul> | None (Unrestricted System Access)                                                                                                                              |
| SALES USER | <ul style="list-style-type: none"><li>• Register new customer inquiries.</li><li>• Build and issue sales quotations.</li><li>• Convert approved quotes to Sales Orders.</li><li>• View stock availability and catalogs.</li></ul>          | <ul style="list-style-type: none"><li>• Cannot dispatch orders.</li><li>• Cannot alter stock baseline figures.</li><li>• Cannot manage system users.</li></ul> |

## 7. Module Breakdown & Specifications

### Module 1: Authentication & Authorization

Provides centralized access control utilizing JSON Web Tokens (JWT) and encrypted password authentication.

- **Login Flow:** User credentials submitted → bcrypt password validation → System signs a JWT populated with user identity and assigned role → Token returned to client for subsequent bearer header requests.

### Module 2: Customer Management

Serves as the master repository for customer profiles, contact vectors, and past interaction histories.

### Module 3: Product Management

Maintains universal product catalogs, pricing indexes, standard Stock Keeping Units (SKUs), and base quantities.

### Module 4 & 5: Enquiry & Quotation Management

Captures client requirement logs and computes commercial offers incorporating dynamic tax calculations, discounts, and itemized line items. *Rule: One enquiry can yield multiple iteration quotations.*

### Module 6: Sales Order Management

Converts client-accepted quotations into binding Sales Orders. Validates that duplicate sales orders are blocked and target products exist.

### Module 7 & 8: Inventory Control & Stock Reservation

Tracks real-time stock allocation to prevent overselling. When an order is confirmed, inventory is locked into reserved\_quantity.

**INVENTORY METRIC FORMULA:**

**Available Stock = Physical Stock – Reserved Stock**

*Example:* If Physical Baseline = 100 units and Reserved = 20 units, Available Stock reflects 80 units for future quotations.

### Module 9: Dispatch Management

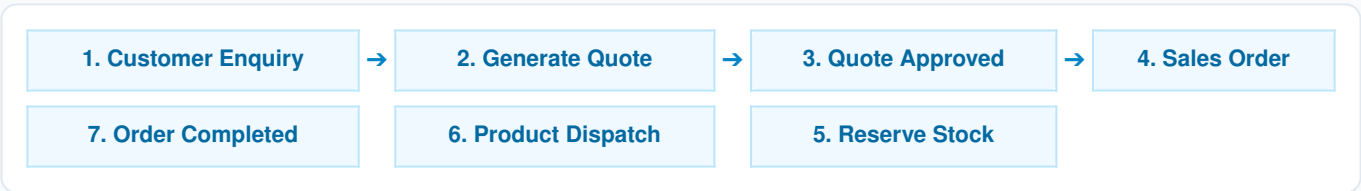
Handles physical product releases. Updating dispatch status automatically decreases physical and reserved stock simultaneously while recording permanent audit entries.

## 8. Database Schema Design

| Table Name | Primary Fields                                        | Key Relationships & Constraints                          |
|------------|-------------------------------------------------------|----------------------------------------------------------|
| Users      | id, name, email, password, role                       | Unique index on email; Enum constraint on role.          |
| Customers  | customer_id, customer_name, email, phone              | Referenced by Enquiries, Quotations, and Sales Orders.   |
| Products   | product_id, name, sku, price, quantity                | Unique SKU index; Foreign key in Inventory.              |
| Inventory  | inventory_id, product_id, available_qty, reserved_qty | FK to Products .product_id. Dynamic transactional locks. |

| Table Name   | Primary Fields                                    | Key Relationships & Constraints                          |
|--------------|---------------------------------------------------|----------------------------------------------------------|
| Enquiries    | enquiry_id, customer_id, enquiry_date, status     | FK to Customers; One-to-Many with Quotations.            |
| Quotations   | quotation_id, enquiry_id, total_amount, status    | FK to Enquiries; Approved quote converts to Sales Order. |
| Sales_Orders | order_id, quotation_id, customer_id, status       | FK to Quotations & Customers; Triggers reservation.      |
| Dispatches   | dispatch_id, order_id, dispatch_date, tracking_no | FK to Sales_Orders; Decrements physical stock.           |

## 9. End-to-End Business Workflow



*Traceability Guarantee:* Every order line item can be audited back to its originating sales quotation and initial customer enquiry log, maintaining full organizational visibility.

## 10. Security Implementation

- **Authentication:** Stateful request authorization handled via signed JWT tokens passed via Auth headers.
- **Password Encryption:** Salted password hashing utilizing bcrypt with configurable cost factors.
- **Role Validation:** Express route-level middleware enforcing permission boundaries before endpoint handler execution.
- **Data Integrity:** Sequelize DB transactions (managed transactions) wrapped around stock reservation and dispatch operations to ensure rollback on failure.

## 11. Quality Assurance & Testing Strategy

| Testing Scope        | Methodology & Target Components                                             | Key Verification Points                                              |
|----------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| Unit Testing         | Isolated function execution (e.g., math utilities, calculation algorithms). | Quotation price totals, discount applying logic, stock math formula. |
| Integration Testing  | Inter-module communication & workflow state transitions.                    | Quotation → Sales Order creation; Order → Reserve Inventory flow.    |
| API Endpoint Testing | HTTP requests to Express endpoints (REST validation).                       | Response status codes, JWT validation, payload schema checks.        |

## 12. Key Advantages

- High operational execution speed over paper/Excel.
- Eliminates double-booking of physical warehouse stock.
- Seamless scale handling due to stateless Express/Node backend.
- Strict transactional data integrity backed by PostgreSQL.

## 13. System Limitations

- Lacks integrated Procurement / Supplier module.
- Does not currently auto-generate GST compliance forms.
- Absence of real-time cloud multi-warehouse syncing.
- Reporting limited to essential transactional data.

## 14. Future Enhancement Roadmap

| Phase                         | Planned Capabilities                                                                      |
|-------------------------------|-------------------------------------------------------------------------------------------|
| Phase 1: Compliance & Docs    | Automated GST PDF invoice generator, email dispatch receipts, SMS notifications.          |
| Phase 2: Warehouse Tech       | Barcode / QR Code scanning integration for instant dispatch verification.                 |
| Phase 3: Intelligence & Scale | AI-driven stock demand forecasting, multi-branch support, AWS Cloud Deployment (EC2/RDS). |

## 15. Conclusion

The **PERN ERP System** provides a robust, scalable, and highly reliable technical solution for enterprise resource management. By combining modern web paradigms—React's interactive rendering, Express/Node's high-performance API processing, and PostgreSQL's atomic database engine—the platform effectively eliminates manual operational flaws, enforces strict inventory tracking, and provides absolute traceability across the entire enterprise sales lifecycle.